Pepsin: Pepsin is secreted in the form of inactive pepsinogen, a protein with a molecular weight of 42 kD. Pepsinogen is cleaved into active pepsin of 34 kD, initially by the acidic medium, and then autocatalytically.

Pepsin

- is an endopeptidase (an enzyme that acts in the interior of the protein)
- has a broad range of specificity
- acts on the peptide linkages contributed by the carboxyl group of aromatic amino acids / acidic amino acids, to a lesser extent.
- breaks down proteins to peptones.
- works optimally at pH 2.0. Hence, any factor that reduces acidity of the stomach like achlorhydria or antacid intake that neutralizes stomach acid or proton pump inhibitors reduces the activity of pepsin.

Rennin: Rennin, also called as chymosin is an enzyme that is responsible for curdling of milk and is present only in infants and children. It is responsible for conversion of milk protein into Paracaseinate.

Gelatinase: Gelatinase helps in the digestion of gelatin and type IV and V collagen, which are proteoglycans present in meat. Gastricsin is another enzyme present in the gastric juice that digests proteins.

Digestion in duodenum

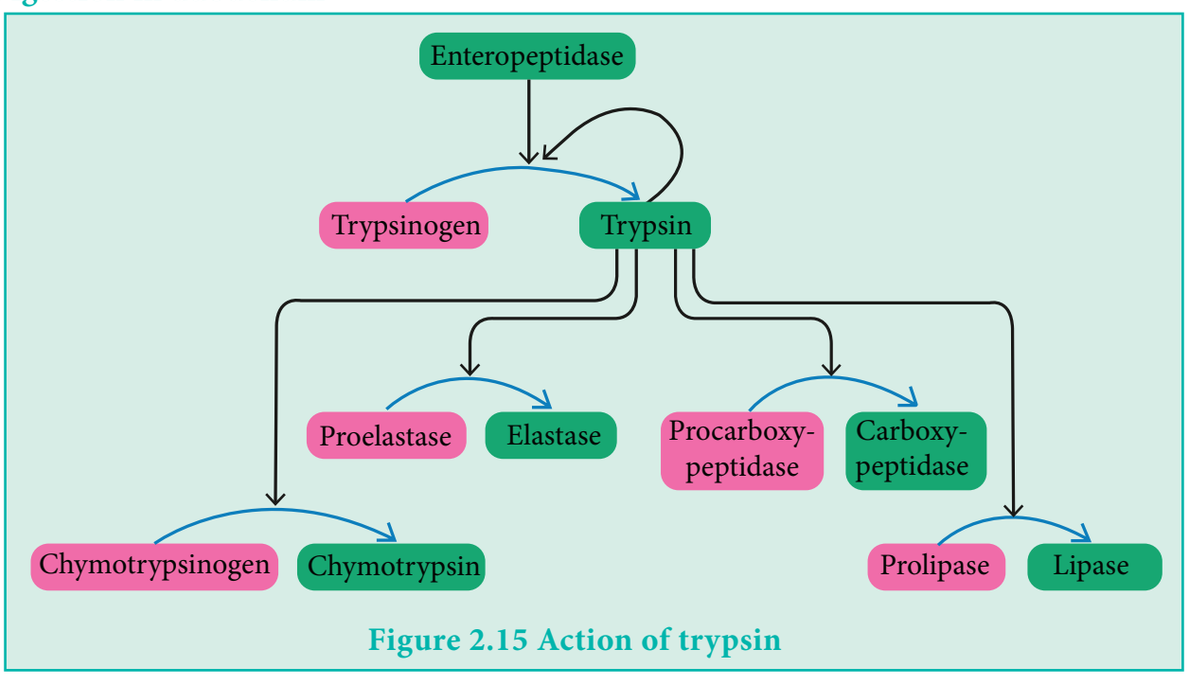


Figure 2.15 Action of trypsin

Enteropeptidase: Enteropeptidase or Enterokinase is a transmembrane protein expressed by the epithelial cells of the jejunum and the duodenum. It has a substrate specificity of Lys-Lys or Arg-Lys and is involved in activation of trypsin.

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